

Artifact Repository Manager with Nexus

Key Takeaways

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Introduction to Artifact Repository Manager



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What is an Artifact Repository?

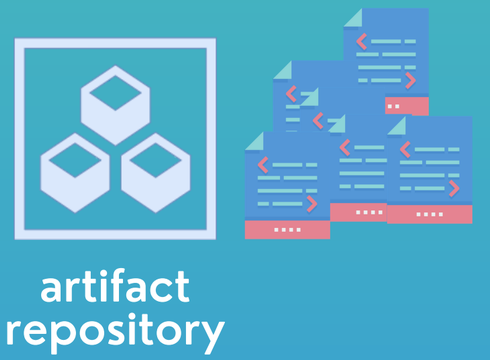
- **Storage of build artifacts** produced by continuous integration which makes them available for automated deployment to different deployment environments
- Provides a **central location**



- Artifacts are applications built into a **single file**
- There are different artifact **formats:**

JAR, WAR, ZIP, TAR, etc.

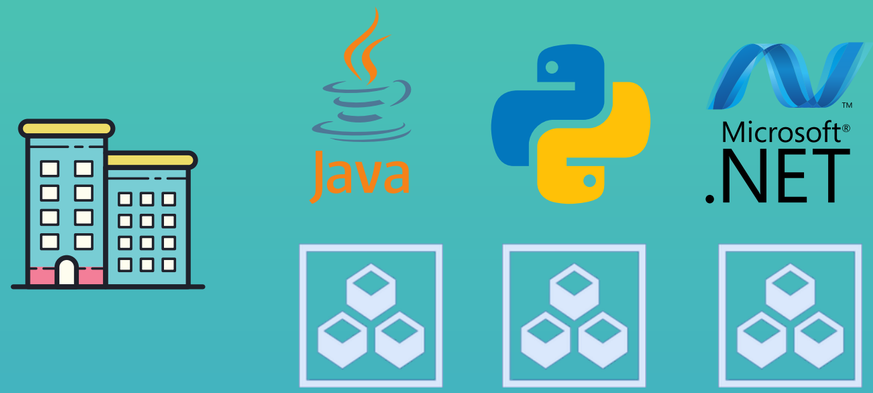
- Artifact repository needs to **support this specific format**
- Repository for each file/artifact type



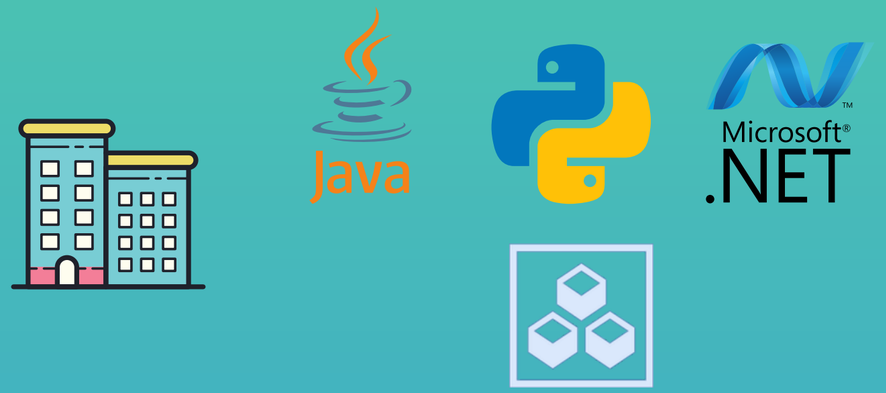
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What is an Artifact Repository Manager?

- Can **store many different artifact types**
- Which is great, because as a company you often produce different types of artifacts



Instead of **having different repositories for each** artifact type



Just 1 repository for managing all your different artifact types

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Public vs Private Artifact Repository Managers

Public Repository Managers

- There are public repository managers
- For example libraries/frameworks you use as a dependency
- You can make own project publically available there:



Public

Nexus - one of the most popular repository managers

- It's a private central store
- You can **upload** and **store** different build artifacts
- **Retrieve** (download) artifacts later
- For internal usage in your company

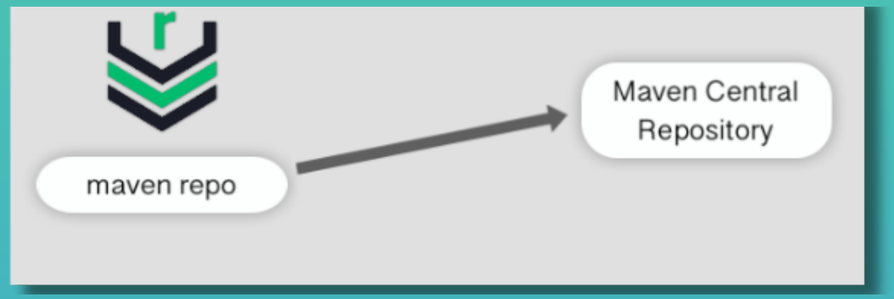
nexus repository

Private

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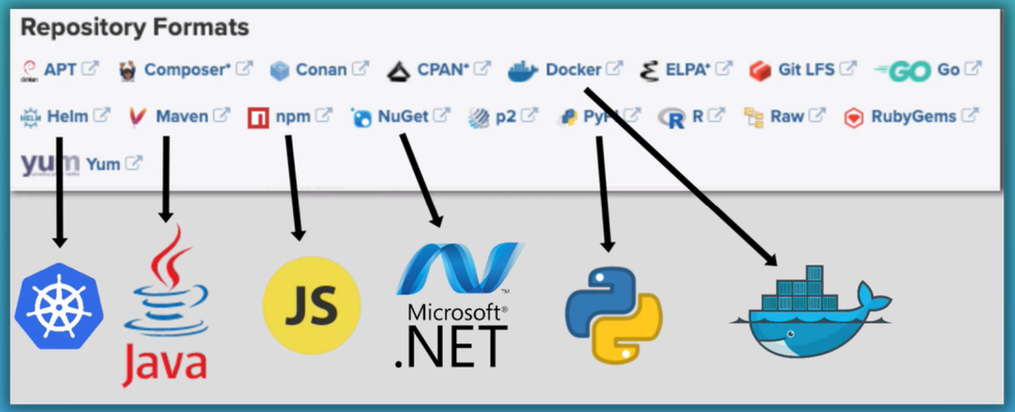
Nexus - Artifact Repository Manager

- Has open source and commercial version
- Host own repositories
- Proxy repositories (intermediary to another repository):



Fetch everything from 1 place (Nexus):
Company internal and public artifacts

- You can have multiple repositories for different formats or different teams in your company
- **Supported formats** in Nexus:

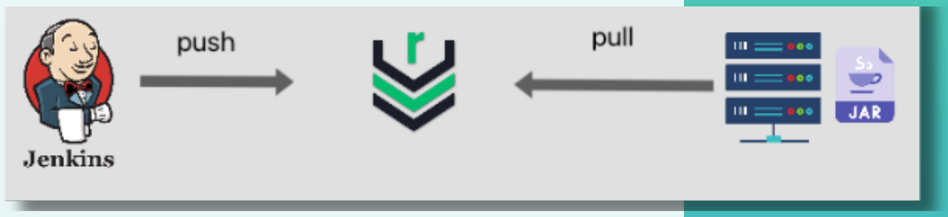


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Features of Repository Manager

- Integration with LDAP
- Flexible and powerful REST API for integration with other tools

Important, because you need to integrate it into your CI/CD Pipeline



- Backup and Restore
- Multi-Format Support (different file types - zip, tar, docker etc)
- Metadata Tagging (labelling and tagging artifacts)
- Cleanup Policies (automatically delete files that match condition)
- Search functionality (across projects, artifact repos etc.)
- User token support for system user authentication

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Install and Run Nexus on a Cloud Server



Install and Run Nexus on a Cloud Server - 1

Summary of Steps - 1st Part:

1. **Create Ubuntu Server (Droplet)** - min 4GB RAM, 2 CPUs, 160 GB SSD Disk
2. **Open SSH port 22**
3. **Install Java**
4. **Download and Install Nexus**
5. **Create "nexus" user and group** - Best Practice: Run applications with own user

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Install and Run Nexus on a Cloud Server - 2

With Nexus installation you get:

```
nana — root@ubuntu-s-4vcpu-8gb-fra1: /opt — ssh root@165.22.86.16 — 118×32
root@ubuntu-s-4vcpu-8gb-fra1:/opt# ls
digitalocean  nexus-3.94.0-12  nexus-3.94.0-12-linux-x86_64.tar.gz  sonatype-work
root@ubuntu-s-4vcpu-8gb-fra1:/opt# ls nexus-3.94.0-12
NOTICE.txt  OSS-LICENSE.txt  bin  deploy  etc  jdk
root@ubuntu-s-4vcpu-8gb-fra1:/opt# ls sonatype-work/nexus3/
clean_cache  log  tmp
root@ubuntu-s-4vcpu-8gb-fra1:/opt#
```

Sonatype-work:
contains own config for Nexus and data

- Subdirectories depending on your Nexus configuration
- IP address that accessed Nexus
- Logs of Nexus App
- Your uploaded files and metadata
- You can use this folder for backup





Nexus folder:
contains runtime and application of Nexus

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Install and Run Nexus on a Cloud Server - 3

Summary of Steps - 2nd Part:

1. Make Nexus owner of the folders
2. Start with nexus user
3. Open port 8081 with firewall
4. Access from browser

Starting Nexus...

- Services should not run with root user permissions
- So best practice: create own user for service (like in this case **nexus**)
- Should only have the permission for that specific service



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Install and Run Nexus on a Cloud Server - 4

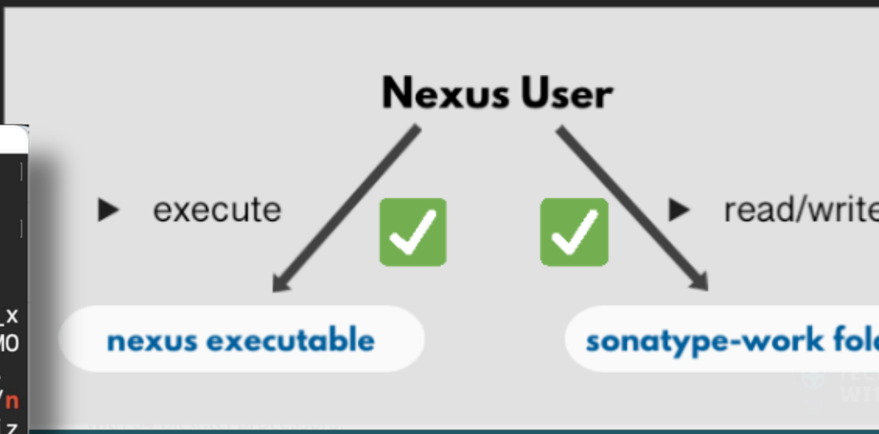
Run Nexus...

1. Make Nexus own the folders

2. Start with nexus user

```
root@ubuntu-s-4vcpu-8gb-fra1:/opt# ls -l
total 517224
drwxr-xr-x 4 root root      4096 Jul 20 14:19 digitalocean
drwxr-xr-x 6 root root      4096 Jul 20 14:20 nexus-3.94.0-12
-rw-r--r-- 1 root root 529617154 Jul  9 15:11 nexus-3.94.0-12-linux-x86_64.tar.gz
drwxr-xr-x 3 root root      4096 Jul  8 15:27 sonatype-work
root@ubuntu-s-4vcpu-8gb-fra1:/opt# chown -R nexus:nexus nexus-3.94.0-12
root@ubuntu-s-4vcpu-8gb-fra1:/opt# chown -R nexus:nexus sonatype-work/
root@ubuntu-s-4vcpu-8gb-fra1:/opt# ls -l
total 517224
drwxr-xr-x 4 root  root       4096 Jul 20 14:19 digitalocean
drwxr-xr-x 6 nexus nexus      4096 Jul 20 14:20 nexus-3.94.0-12
-rw-r--r-- 1 root  root  529617154 Jul  9 15:11 nexus-3.94.0-12-linux-x86_64.tar.gz
drwxr-xr-x 3 nexus nexus      4096 Jul  8 15:27 sonatype-work
root@ubuntu-s-4vcpu-8gb-fra1:/opt#
```

```
nexus@ubuntu-s-4vcpu-8gb-fra1:~$ /opt/nexus-3.94.0-12/bin/nexus start
Starting nexus
nexus@ubuntu-s-4vcpu-8gb-fra1:~$ ps aux | grep nexus
root        1109  0.0  0.0   9940  4508 pts/0    S    14:36   0:00 su - nexus
nexus       1110  0.1  0.0   9060  5440 pts/0    S    14:36   0:00 -bash
nexus       1366  209 24.0 6210508 1959032 pts/0 Sl   14:36   0:52 /opt/nexus-3.94.0-12/jdk/temurin_21.0.11_10_linux_x
86_64/jdk-21.0.11+10/bin/java -server -Dnexus.installer.type=linux-x86-64 -Xms2703m -Xmx2703m -XX:+UnlockDiagnosticVMO
ptions -XX:+LogVMOutput -XX:LogFile=../sonatype-work/nexus3/log/jvm.log -XX:-OmitStackTraceInFastThrow -Dkaraf.home=.
-Dkaraf.base=. -Djava.util.logging.config.file=etc/spring/java.util.logging.properties -Dkaraf.data=../sonatype-work/n
exus3 -Dkaraf.log=../sonatype-work/nexus3/log -Djava.io.tmpdir=../sonatype-work/nexus3/tmp -Djdk.tls.ephemeralDHKeySiz
e=2048 -Dfile.encoding=UTF-8 --add-reads=java.xml=java.logging --add-opens java.base/java.security=ALL-UNNAMED --add-o
pens java.base/java.net=ALL-UNNAMED --add-opens java.base/java.lang=ALL-UNNAMED --add-opens java.base/java.util=ALL-UN
NAMED --add-opens java.naming/javax.naming.spi=ALL-UNNAMED --add-opens java.rmi/sun.rmi.transport.tcp=ALL-UNNAMED --ad
d-exports=java.base/sun.net.www.protocol.http=ALL-UNNAMED --add-exports=java.base/sun.net.www.protocol.https=ALL-UNNAM
ED --add-exports=java.base/sun.net.www.protocol.jar=ALL-UNNAMED --add-exports=jdk.xml.dom/org.w3c.dom.html=ALL-UNNAMED
 --add-exports=jdk.naming.rmi/com.sun.jndi.url.rmi=ALL-UNNAMED --add-exports=java.security.sasl/com.sun.security.sasl=
ALL-UNNAMED --add-exports=java.base/sun.security.x509=ALL-UNNAMED --add-exports=java.base/sun.security.rsa=ALL-UNNAMED
 --add-exports=java.base/sun.security.pkcs=ALL-UNNAMED -jar /opt/nexus-3.94.0-12/bin/sonatype-nexus-repository-3.94.0-
12.jar
nexus       1399  100  0.0  11320  4456 pts/0    R+   14:36   0:00 ps aux
nexus       1400  0.0  0.0   7076  2164 pts/0    S+   14:36   0:00 grep --color=auto nexus
nexus@ubuntu-s-4vcpu-8gb-fra1:~$
```



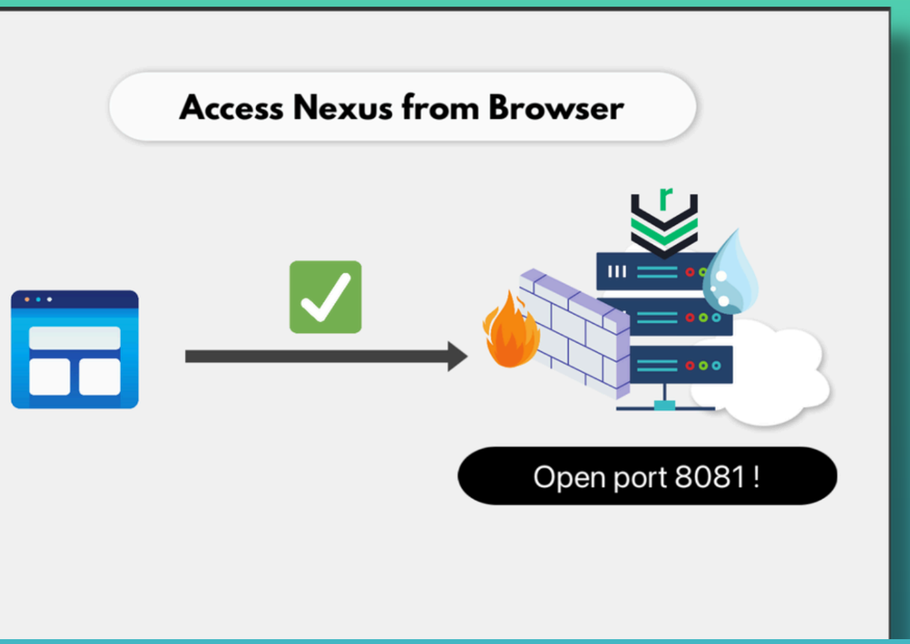
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Install and Run Nexus on a Cloud Server - 5

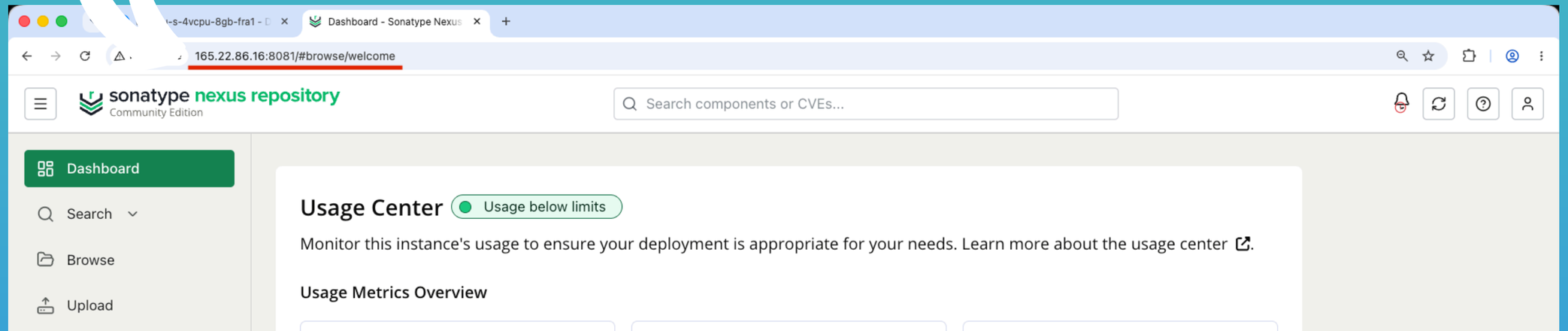
Run Nexus...

3. Open port 8081 on Firewall

```
nexus@ubuntu-s-4vcpu-8gb-fra1:~$ /opt/nexus-
Starting nexus
nexus@ubuntu-s-4vcpu-8gb-fra1:~$ ps aux | gr
root        1109  0.0  0.0   9940  4508 pts/
nexus       1110  0.1  0.0   9060  5440 pts/
nexus       1366  209 24.0 6210508 1959032 p
86_64/jdk-21.0.11+10/bin/java -server -Dnexu
ptions -XX:+LogVMOutput -XX:LogFile=../sonat
-Dkaraf.base=. -Djava.util.logging.config.fi
exus3 -Dkaraf.log=../sonatype-work/nexus3/lo
=2048 -Dfile.encoding=UTF-8 --add-reads=jav
ens java.base/java.net=ALL-UNNAMED --add-op
AMED --add-opens java.naming/javax.naming.s
d-exports=java.base/sun.net.www.protocol.ht
ED --add-exports=java.base/sun.net.www.proto
 --add-exports=jdk.naming.rmi/com.sun.jndi.u
ALL-UNNAMED --add-exports=java.base/sun.secu
 --add-exports=java.base/sun.security.pkcs=A
12.jar
nexus       1399  100  0.0  11320  4456 pts/
nexus       1400  0.0  0.0   7076  2164 pts/
nexus@ubuntu-s-4vcpu-8gb-fra1:~$ netstat -lr
(Not all processes could be identified, non-
 will not be shown, you would have to be roo
```



4. Access from browser



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Overview of Nexus

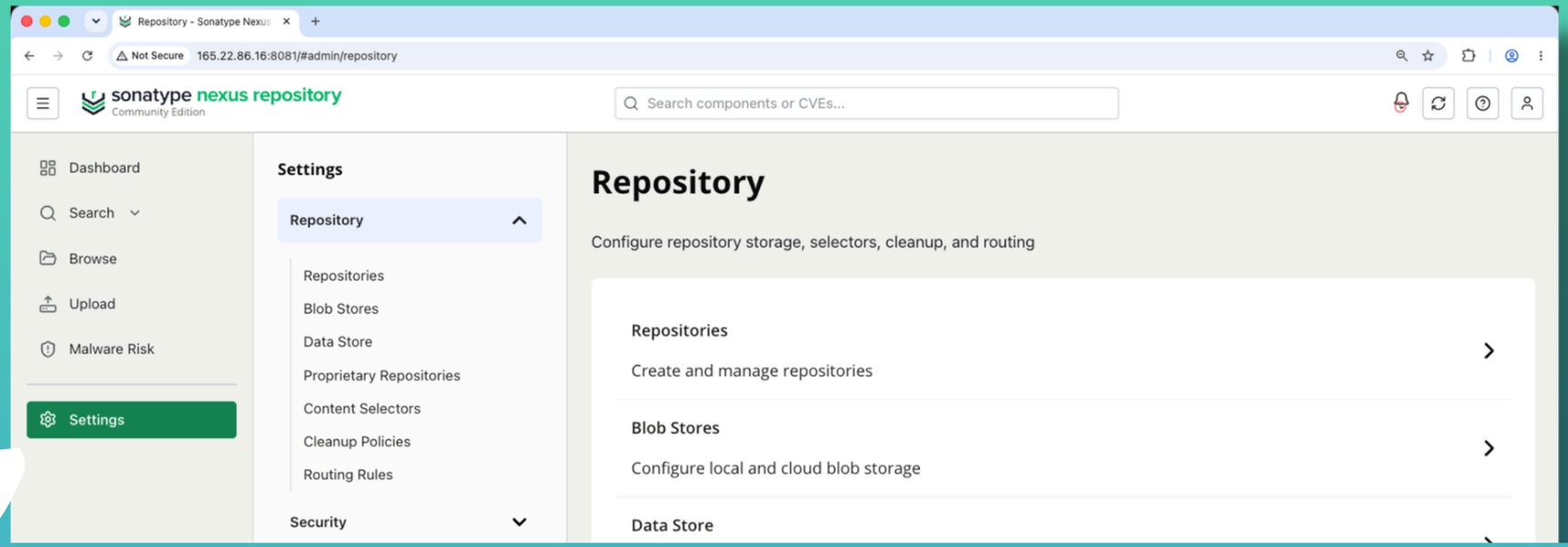


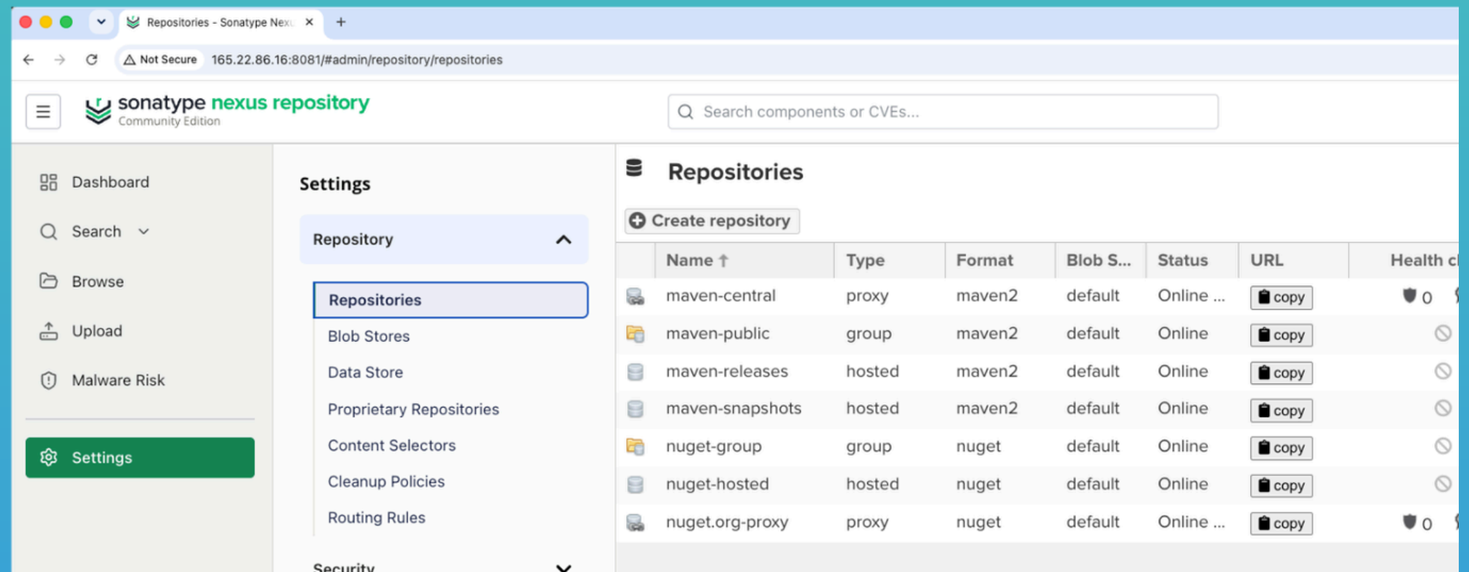
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Login & Default Repositories

Login

- With **default "admin" user** that automatically gets created to administer Nexus





Default Repositories

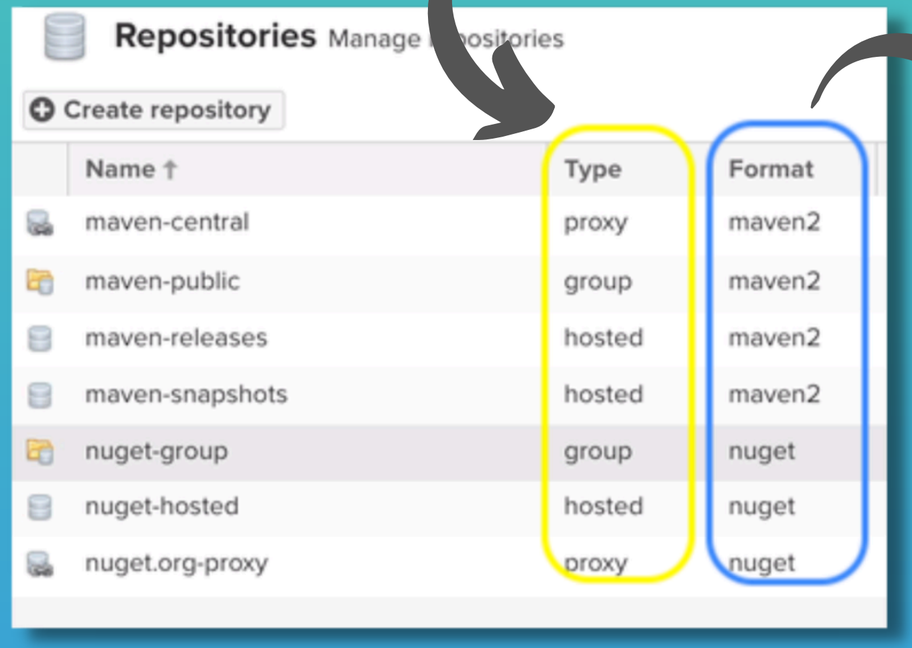
- You get default repositories with different **types** when you install Nexus

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Repository Types & Formats

Repository Types

- **Proxy**: intermediary to another repository
- **Group**: containing a group of repositories
- **Hosted**: for hosting company-internal artifacts





Each technology (python, java, docker) has its artifact type (jar, image), which can be saved in its own artifact repository format

Repository Formats

- **maven2**: for java jar artifacts
- **nuget**: for .NET packages
- **docker**: for Docker images

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Publish artifact to repository

You can push or "PUBLISH" artifacts to Nexus with different tools

- Using built-in commands of build tools, like Maven or Gradle
- For that, you need to **configure** each build tool **with Nexus credentials and address**





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Nexus REST API

You can also **interact with Nexus using its REST API** to do multiple things:

- query for different information: list of repositories | list of components | ...
- download an artifact
- upload an artifact

How to interact?

- Using a tool like curl or wget to execute http request
- You need to provide username and credentials of a Nexus user

```
nana@macbook /Users/nana
% curl -u user:pwd -X GET 'http://157.230.22.19:8081/service/rest/v1/repositories'
```

```
nana@macbook /Users/nana
% curl -u nana:techworld -X GET 'http://157.230.22.19:8081/service/rest/v1/components?repository=maven-snapshots'
```



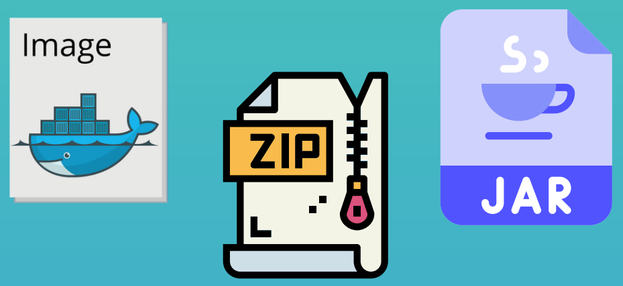
To use Nexus API or any API:
You should **always reference the documentation**. Not learn by heart, because 1) API's change 2) there are a lot of endpoints

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Component vs Asset

Components

- **abstract**
- what we are uploading
- term "component" refers to any type or format



Assets

- **actual physical packages/files**
- 1 component = 1 or more assets

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Cleanup Policies & Scheduled Tasks

Cleanup Policies

- Clean up old artifacts to make space. You can define logic like:

→ **Delete all artifacts older than 30 days**

→ **Delete all artifacts, which haven't been used/downloaded for more than 10 days**



- You can create individual cleanup policy for each repository

Scheduled Tasks

- Nexus allows you to schedule tasks that will be applied to all or a specific repository on a configurable schedule

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